

FILM, Sweden

WMO: 024560

Lat: 60.2358N

Lon: 17.9043E

Elev: 115

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.6	4.5	-5.5	4.1	-0.9	1.0	5.8	6.7	16.3	38.3	14.5	36.3	0.7	210	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	80.8	65.5	77.3	63.8	74.0	62.0	67.8	76.2	65.7	74.1	63.8	71.4	6.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	92.0	71.3	62.7	85.6	68.9	60.7	79.7	67.1	32.2	76.0	30.6	74.0	29.2	71.5	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	12.0	10.6	-7.8	85.0	7.1	3.1	-13.0	87.2	-17.1	89.0	-21.1	90.8	-26.3	93.0
			-8.5	69.5	7.2	2.7	-13.7	71.4	-17.9	73.0	-21.9	74.5	-27.1	76.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.5	25.7	25.9	31.0	40.2	48.9	56.5	62.3	60.3	52.2	42.2	34.8	29.3
	DBStd	14.74	9.59	9.34	8.37	6.81	6.77	5.85	5.44	5.26	5.90	6.66	6.93	9.07
	HDD50	3837	753	676	589	303	104	11	0	2	46	255	457	641
	HDD65	8264	1218	1096	1054	745	499	261	121	164	385	708	907	1106
	CDD50	1109	0	0	0	8	71	207	380	320	111	12	0	0
	CDD65	60	0	0	0	0	1	6	36	17	0	0	0	0
	CDH74	700	0	0	0	0	23	109	405	161	2	0	0	0
	CDH80	117	0	0	0	0	2	12	84	19	0	0	0	0
	WSAvg	4.7	4.7	4.7	5.0	5.2	5.3	4.7	4.4	4.1	4.4	4.6	4.8	4.7
	PrecAvg	24.4	1.8	1.5	1.3	1.5	1.6	2.5	2.6	2.9	2.2	2.2	2.4	2.0
Precipitation	PrecMax	34.5	3.4	3.0	2.4	4.4	3.5	5.2	5.1	6.1	5.0	4.8	5.0	4.0
	PrecMin	19.1	0.2	0.3	0.1	0.2	0.6	0.4	0.7	0.2	0.6	0.2	0.6	0.5
	PrecStd	3.1	0.8	0.6	0.6	1.0	0.7	1.3	1.4	1.4	1.1	1.1	1.1	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.8	46.2	56.2	68.6	77.1	81.6	86.3	82.3	72.8	61.1	52.4	50.1
		MCWB	40.9	40.9	45.4	53.0	60.0	63.9	66.5	66.9	61.4	55.1	50.1	46.9
	2%	DB	41.5	42.6	50.5	62.9	71.9	77.1	82.3	78.5	68.9	57.2	49.3	45.8
		MCWB	39.2	38.4	41.9	49.6	57.1	61.4	66.4	65.3	59.8	53.2	47.4	43.7
	5%	DB	39.6	39.9	46.4	58.0	67.4	73.4	78.8	74.9	65.4	54.6	46.9	43.2
		MCWB	37.7	37.1	40.1	47.3	54.7	59.5	65.4	63.4	58.3	51.5	45.2	41.4
	10%	DB	37.4	37.4	42.6	53.4	63.1	69.7	74.8	71.8	62.4	52.3	44.8	40.7
		MCWB	36.0	35.2	37.6	44.3	52.4	58.2	63.6	62.3	56.5	49.7	43.4	39.3
	0.4%	WB	42.1	42.4	46.1	54.5	62.4	66.3	70.5	70.7	63.0	56.4	50.4	47.4
		MCDB	43.3	45.3	54.3	66.6	73.2	77.5	79.3	78.1	69.9	59.6	51.6	49.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.6	39.3	43.0	50.5	58.8	63.4	68.6	67.2	60.9	53.9	47.7	44.1
		MCDB	40.9	41.5	49.4	60.6	68.5	73.5	77.6	74.4	66.5	56.3	49.1	45.4
	5%	WB	37.8	37.5	40.6	48.1	55.9	61.2	66.6	65.2	59.1	52.1	45.5	41.5
		MCDB	39.1	39.7	45.2	57.0	65.6	70.5	75.6	72.1	64.3	54.3	46.7	42.7
	10%	WB	36.0	35.5	38.4	45.4	53.3	59.2	64.8	63.3	57.4	50.0	43.6	39.2
		MCDB	37.3	37.2	42.2	52.2	61.6	67.6	73.1	69.6	61.5	51.9	44.8	40.5
Mean Daily Temperature Range	5% DB	MDBR	9.8	12.1	15.2	18.7	19.8	19.2	19.5	18.8	16.8	12.6	8.4	8.9
		MCDBR	8.9	13.2	20.3	27.1	28.3	26.8	25.5	23.9	20.7	15.0	10.5	9.5
		MCWBR	8.1	10.1	13.0	15.5	15.8	13.5	12.0	12.0	12.6	11.3	9.5	8.6
	5% WB	MCDBR	8.3	11.3	18.7	24.7	24.7	22.8	21.8	19.9	18.5	13.2	10.2	9.1
		MCWBR	7.9	9.0	12.6	14.7	14.7	12.7	11.3	11.2	11.9	10.5	9.5	8.5
Clear-Sky Solar Irradiance	taub		0.271	0.313	0.333	0.367	0.366	0.357	0.382	0.374	0.341	0.328	0.295	0.380
	taud		2.200	2.284	2.355	2.341	2.405	2.462	2.411	2.432	2.509	2.426	2.243	2.675
	Ebn at Noon		158	216	252	263	273	278	266	259	250	211	145	113
	Edh at Noon		13	21	27	33	33	32	33	30	23	18	12	9
All-Sky Solar Radiation	RadAvg		85	277	751	1241	1629	1815	1684	1317	842	372	111	54
	RadStd		11	49	87	117	150	122	166	126	96	60	26	6

Historical Trends

DBAvg	Heating		Cooling			Degree-Days			
	99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	-198	N/A	N/A	N/A

Nomenclature: See separate page